

Candidates' Performance

Paper 1

Paper 1 consists of two sections: multiple-choice questions in Section A and conventional questions in Section B. All questions in both sections are compulsory.

Section A (multiple-choice questions)

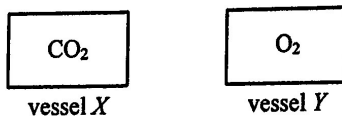
Section A consisted of 33 multiple-choice questions and the mean score was 19. Post-examination analysis revealed the following:

1. An equal amount of energy is supplied to each of the substances *P*, *Q*, *R* and *S* of mass and specific heat capacity listed below. Which substance would have the largest rise in temperature?

	substance	mass / kg	specific heat capacity / J kg ⁻¹ °C ⁻¹	
A.	<i>P</i>	2	4200	(6%)
B.	<i>Q</i>	3	2100	(1%)
*C.	<i>R</i>	4	900	(88%)
D.	<i>S</i>	5	840	(5%)

Most candidates correctly applied the relation $\Delta T = \frac{Q}{mc}$, identifying that substance *R*, having the smallest product of mass and specific heat capacity, would undergo the largest temperature rise.

3. Identical closed vessels *X* and *Y* contain carbon dioxide (CO₂) and oxygen (O₂) respectively.



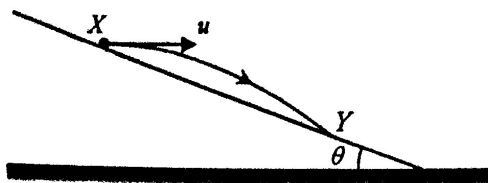
Both gases are at room temperature and atmospheric pressure. Which of the following physical quantities of the gas molecules in the two vessels is/are the same?

- (1) root-mean-square speed
- (2) average kinetic energy
- (3) frequency of collision with the walls of the vessel

- | | | |
|-----|------------------|-------|
| A. | (1) only | (11%) |
| *B. | (2) only | (33%) |
| C. | (1) and (3) only | (13%) |
| D. | (2) and (3) only | (43%) |

About half of the candidates did not realise that the frequency of collision of the molecules with the walls of the vessel should be different for the two gases.

9.

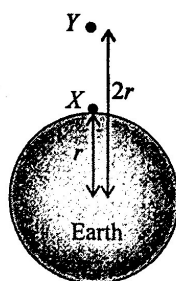


X and Y are two points on an incline making an angle θ with the horizontal. A particle projected horizontally from X with speed u lands at Y as shown. Which expression correctly gives its time of flight from X to Y ? Neglect air resistance.

- | | | |
|-----|----------------------------|-------|
| A. | $\frac{u \sin \theta}{2g}$ | (15%) |
| B. | $\frac{u \tan \theta}{2g}$ | (19%) |
| C. | $\frac{2u \sin \theta}{g}$ | (18%) |
| *D. | $\frac{2u \tan \theta}{g}$ | (48%) |

Nearly half of the candidates were able to apply the equations of projectile motion and to carry out multi-step calculations to arrive at the correct answer.

12.

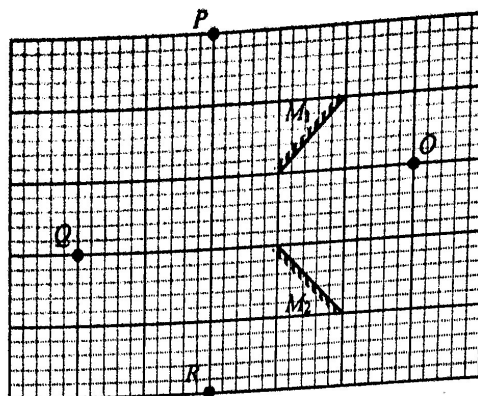


Two identical objects located at point X and point Y are at distances r and $2r$ from the Earth's centre. The weight of the object at point X is W . What is the weight of the object at point Y ?

- | | | |
|-----|---------|-------|
| *A. | $0.25W$ | (77%) |
| B. | $0.5W$ | (13%) |
| C. | $0.75W$ | (4%) |
| D. | W | (6%) |

Over three-quarters of candidates correctly applied the inverse square law of gravitation, recognising that weight decreases with the square of the distance from the Earth's centre.

13. A point object O is placed in front of two plane mirrors M_1 and M_2 as shown.

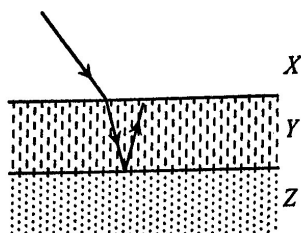


At which position(s) can an image be formed?

- | | | |
|-----|------------------|-------|
| A. | P, Q and R | (19%) |
| *B. | Q and R only | (25%) |
| C. | R only | (47%) |
| D. | P only | (9%) |

Candidates who chose option C identified only the image (R), which is formed by a single reflection and failed to identify the additional image (Q), which results from multiple reflections.

14.



The figure shows three media X, Y and Z separated by parallel boundaries. A light ray travelling from X into Y then undergoes total internal reflection at the boundary between Y and Z . Arrange in descending order the speed of light in the respective media.

- | | | |
|-----|---------|-------|
| A. | $X Y Z$ | (19%) |
| B. | $X Z Y$ | (18%) |
| C. | $Y Z X$ | (13%) |
| *D. | $Z X Y$ | (50%) |

Many candidates failed to relate the speed of light in various media to the bending of the light ray.

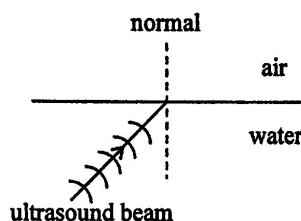
18. In an experiment, monochromatic light is incident normally on a double slit and a diffraction grating placed in turn at the same position. The respective interference and diffraction patterns formed on the screen are observed. Which of the following statements about the patterns is/are correct ?

- (1) The fringe separation of the interference pattern is wider.
- (2) The fringes of the diffraction pattern are sharper.
- (3) The width of the whole diffraction pattern is wider.

- A. (1) only (18%)
- B. (2) only (22%)
- C. (1) and (3) only (23%)
- *D. (2) and (3) only (37%)

Candidates who chose options A and C seemingly did not realise that the pattern produced by a double slit is narrower than that produced by a diffraction grating.

20. A source in water emits a beam of ultrasound towards the water-air interface as shown.



Which of the following statements is **INCORRECT** ?

- A. The frequency of the refracted beam is the same as that of the incident beam. (11%)
- B. The speed of the refracted beam becomes smaller. (16%)
- C. The refracted beam leaving the interface bends towards the normal. (27%)
- *D. If the angle of incidence is large enough, total internal reflection may occur. (46%)

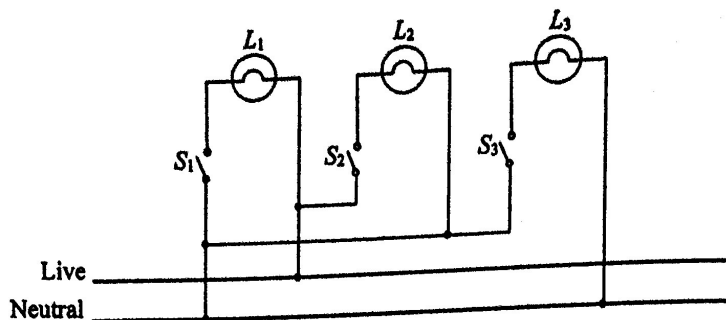
Over half of the candidates incorrectly assumed that total internal reflection could occur when sound travels from water to air, without realising that sound waves travel faster in water.

22. Three light conducting spheres are suspended by nylon threads. Any two of them are found to attract each other when they are brought close to each other. Which conclusion below is correct ?

- A. All three spheres are charged. (8%)
- B. Only one sphere is charged. (7%)
- C. Only one sphere is uncharged and the other two carry like charges. (7%)
- *D. Only one sphere is uncharged and the other two carry unlike charges. (78%)

Nearly 80% of the candidates correctly recognised that the attraction between spheres can occur due to unlike charges and also between a neutral sphere and any charged sphere.

27. The figure shows how lamps L_1 , L_2 and L_3 are connected in the ring circuit of a house. There are some wrong connections in the circuit.



If switches S_2 and S_3 are closed while switch S_1 remains open, which of the lamps will light up?

- | | | |
|-----|----------------------|-------|
| *A. | L_2 only | (33%) |
| B. | L_3 only | (14%) |
| C. | L_2 and L_3 only | (43%) |
| D. | none of them | (10%) |

Candidates who chose option B or C seemingly failed to recognise that there is no potential difference across L_3 even if S_3 is closed.

30. An electric iron works at its rated power of 1100 W under a 50 Hz a.c. mains of root-mean-square voltage 220 V. What power will it deliver when working under a 100 Hz a.c. of root-mean-square voltage 110 V?

- | | | |
|-----|--------|-------|
| *A. | 275 W | (48%) |
| B. | 778 W | (12%) |
| C. | 550 W | (22%) |
| D. | 1100 W | (18%) |

Nearly half of the candidates correctly applied the relation $P \propto V_{rms}^2$, demonstrating that they were aware that power is independent of frequency.

Section B (conventional questions)

Question Number	Performance in General
1	This question tested candidates' understanding of heat transfer and electromagnetic radiation emitted from hot objects. Candidates' performance was good. In (a), about two thirds recognised that thermal energy from the water is transferred to the cup, while some incorrectly attributed the heat loss to evaporation. Part (b) was well attempted. In (c), a high proportion showed that they knew that hot objects emit infra-red radiation. However, about a third failed to name a relevant daily-life device.
2	This question assessed candidates' application of kinematic concepts to projectile motion. The overall performance was fair. Part (a) was generally well handled, though some only calculated the time to reach the highest point. Others overlooked the opposing directions of initial velocity and acceleration. In (b), a number of candidates were apparently unaware of the horizontal velocity at the highest point and incorrectly claimed that the kinetic energy there is zero. Responses to (c) were poor; most did not mention that the horizontal velocity of the two objects is identical, and many mistakenly argued that the two objects travel the same distance.
3	This question required candidates to interpret a motion graph. Candidates' performance was satisfactory. In (a), about a quarter provided a complete description of the motion. Performance in (b) was satisfactory, although some did not specify the position with reference to point O. Part (c) was well answered.
4	This question tested candidates' understanding of circular motion. The overall performance was fair. About two thirds determined the angular speed correctly in (a). Part (b) was poorly answered, with some incorrectly naming the force as 'centripetal force'. Just under half gave a full explanation in (c), while a few omitted that the angular speed of Q remained constant.
5	This question assessed candidates' understanding of experimental design and force analysis during a collision. The performance was poor. While most recognised the action-reaction pair between A and B, many failed to draw the force-time graphs correctly.
6	This question tested candidates' knowledge of geometrical optics, focusing on the properties of lenses and practical adjustments. Candidates performed well. Part (a) was well answered, though a few incorrectly described the image as erect. About two thirds answered (b)(i) and (ii) correctly. Over 70% found the focal length using the lens formula, despite errors in the ray diagrams. Part (c) was poorly answered, revealing their limited understandings of the relationship between image and object distances. Slightly under half answered (d) correctly.
7	This question examined candidates' understanding of circuit behaviour and interpretation of experimental data. The overall performance was good. Most answered (a) correctly. In (b), a high percentage calculated the total resistance accurately, though some only noted the difference (0.3Ω) without explaining its cause.
8	This question tested candidates' understanding of electromagnetic induction via rotating coils in a magnetic field. The overall performance was satisfactory. About a third drew a waveform with correct shape, period and phase in (a). Part (b) was well answered, although some wrongly applied $B = \frac{\mu_0 I}{2\pi r}$ to determine the magnetic flux.
9	This question assessed candidates' application of the ideal gas law. The overall performance was good. Some candidates confused the number of moles with the number of molecules when applying the equation.

10	This question tested candidates' understanding of stationary wave and its application to musical instruments. Candidates performed poorly. In (a)(i), about a third labelled some antinodes and nodes, only a third labelled all correctly. Part (a)(ii) was well answered, although a small number seemingly did not understand the concept of 'phase'. Only about a third answered (a)(iii) correctly. In (a)(iv), many failed to determine the wavelength of the next stationary wave pattern. Nearly half drew the correct pattern in (b)(i). In (b)(ii), only a very small number realised that sound quality depends on the superposition of harmonics. The performance in (b)(iii) was poor. Many did not compare appropriate wave characteristics and some mismatched terms like 'stationary' and 'longitudinal'.
11	This question tested candidates' understanding of the efficiency of energy conversion and the power loss in long-distance transmission of electricity. Candidates' performance was satisfactory. Part (a) was well answered, with a very high proportion obtaining the correct answer in (a)(i), and about two thirds completing the calculations in (a)(ii) and (iii). A small number made errors with prefixes of the units. Part (b)(i) was well answered. In (b)(ii), just under half determined the power loss correctly. The performance in (b)(iii) was poor. Some omitted the use of a transformer, while others failed to explain the benefit of high-voltage transmission.
12	This question required candidates to apply the principles of radioactive decay and their understandings of the motion of ions in a magnetic field. The overall performance was satisfactory. In (a)(i), about two thirds wrote a correct equation, although some mistakenly argued that a neutron is emitted during the decay. Around 80% found the decay constant in (a)(ii). Performance in (a)(iii) was also satisfactory, though some confused the count rate of the ancient wood sample with that of the fresh wood. About two thirds correctly stated the meaning of isotopes in (b)(i), and about half sketched the correct path in (b)(ii). The performance in (b)(iii) was poor. Many did not mention that the magnetic force is perpendicular to the velocity.

Paper 2

Paper 2 consists of four sections. Each section contains eight multiple-choice questions and one structured question which carries 10 marks. Section A contains questions on 'Astronomy and Space Science', Section B on the 'Atomic World', Section C on 'Energy and Use of Energy' and Section D on 'Medical Physics'. Candidates were required to attempt all questions in two of the four sections.

Section	A	B	C	D
Popularity	17%	66%	86%	31%

Section A : Astronomy and Space Science

Multiple-choice questions

Post-examination analysis revealed the following:

- 1.5 Stars X and Y have the same surface temperature. The luminosities of X and Y are $L_X = 121000 L_S$ and $L_Y = 0.1 L_S$ respectively, where L_S is the luminosity of the Sun. Find $R_X : R_Y$, where R_X and R_Y are radii of X and Y respectively.

- A. 1210000 : 1 (10%)
- B. 1 : 1210000 (9%)
- *C. 1100 : 1 (67%)
- D. 1 : 1100 (14%)

Over two-thirds of candidates correctly applied the relation $L \propto R^2 T^4$, and successfully manipulated the ratios to obtain the correct answer.

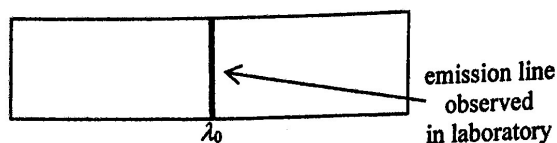
- 1.7 In the Hertzsprung-Russell (H-R) diagram, stars of the same apparent magnitude occupy

- A. the same horizontal line. (28%)
- B. the same vertical line. (14%)
- C. the same diagonal going from lower-right to upper-left. (27%)
- *D. arbitrary locations in the diagram. (31%)

Just under one-third of the candidates correctly recognised that stars of the same apparent magnitude can occupy arbitrary locations in the H-R diagram.



Which of the following best shows the observed emission line (rest wavelength λ_0 observed in laboratory) of the above blueshifted spiral galaxy?



- increasing wavelength
→
- A. (41%)
- B. (15%)
- *C. (30%)
- D. (14%)

The majority of candidates showed that they understood that blueshift causes the wavelength to become shorter. However, less than half of these candidates recognised that the emission line spans a range of wavelengths due to the spiral motion of the galaxy.

Structured questions

Question 1

This question tested candidates' understanding of orbital motions. The overall performance of the candidates was fair. Part (a) was well answered, although a few mistakenly treated the lunar radius as the orbital radius of the orbiting vehicle. Candidates performed poorly in (b) and (c). In (b), many mistakenly argued that the weight of the astronaut is balanced by other forces or that the net force acting on the astronaut is zero. In (c)(i), less than half correctly applied the law of conservation of energy. Some were seemingly confused by the positive and negative signs in the equation, and others omitted the kinetic energy of S_A after docking with S_B . In (c)(ii), more than half recognised that the design could save fuel, though some gave irrelevant answers.

Section B : Atomic World

Multiple-choice questions

Post-examination analysis revealed the following:

- 2.1 In Thomson's plum pudding model, it is suggested that an atom is a positively charged sphere with electrons distributed evenly within it. Which of the following results of the α -scattering experiment implies that Thomson's model is incorrect ?

- (1) Most of the α particles pass straight through the thin gold foil.
- (2) Some α particles are deflected by large angles.
- (3) A few α particles are bounced back.

- A. (1) only (25%)
- B. (2) only (14%)
- C. (1) and (3) only (22%)
- *D. (2) and (3) only (39%)

Candidates who chose options A and C seemingly did not realise that most α particles passing straight through the foil is consistent with both the plum pudding model and the nuclear model, and therefore this does not invalidate Thomson's model.

- 2.2 Which of the following about the Bohr atomic model is/are correct ?

- (1) Electrons are orbiting around the nucleus.
- (2) The attractive force between the nucleus and the electron provides the centripetal force for the electron's circular motion.
- (3) Photons are emitted when electrons jump from lower energy orbits to higher energy orbits.

- A. (2) only (4%)
- B. (3) only (5%)
- *C. (1) and (2) only (78%)
- D. (1) and (3) only (13%)

Nearly 80% of the candidates showed that they have a basic understanding of the motion of electrons in the Bohr atomic model.

- 2.8 When a material is broken up into nanosized particles, which of the following statements is/are correct ?

- (1) Its chemical properties change but the physical properties remain unchanged.
- (2) Its shape changes but the atomic arrangement remains unchanged.
- (3) Its total surface area to volume ratio increases.

- A. (1) only (4%)
- *B. (3) only (39%)
- C. (2) and (3) only (44%)
- D. (1), (2) and (3) (13%)

Candidates who chose options C and D incorrectly assumed that the atomic arrangement remains unchanged, which is not necessarily true for nanomaterials.

Structured questions

Question 2

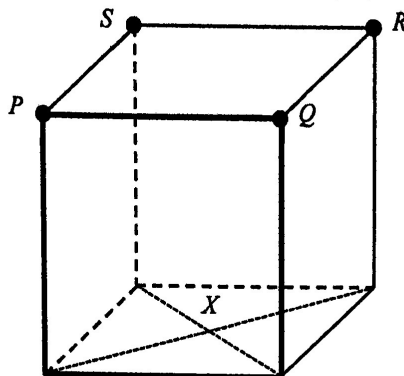
This question required candidates to solve problems involving energy levels of hydrogen atom and the photoelectric effect. The overall performance of candidates was fair. Part (a) was satisfactorily answered. In (a)(i), excitation was commonly mentioned, but few addressed the electron's return to a lower energy level. In (a)(ii), many candidates correctly determined the wavelength of photon. In (b)(i), more than half completed the calculation accurately, although some made mistakes in converting units. Candidates performed poorly in (b)(ii). Most did not label the intercept on the E_p axis, and many incorrectly assumed a linear relationship between v_{max} and E_p . Part (c) was also poorly answered. Many did not use valid values to compare the energy required with the energy of the photon.

Section C : Energy and Use of Energy

Multiple-choice questions

Post-examination analysis revealed the following:

- 3.1 In a room of dimensions $d \times d \times d$, identical point light sources, P , Q , R and S , are located at the four corners of the ceiling as shown. The luminous flux of each source is Φ .

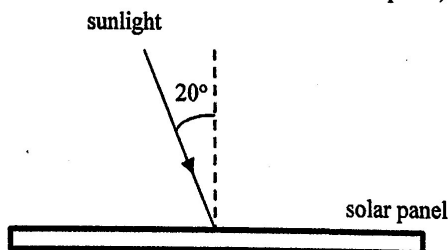


Find the illuminance around point X on the floor in the middle of the room. Neglect reflections from the walls of the room.

- A. $\frac{\Phi}{\pi d^2} \times \left(\frac{3}{2}\right)^{\frac{1}{2}}$ (19%)
 B. $\frac{\Phi}{\pi d^2} \times \left(\frac{3}{2}\right)^{\frac{3}{2}}$ (18%)
 C. $\frac{\Phi}{\pi d^2} \times \left(\frac{2}{3}\right)^{\frac{1}{2}}$ (23%)
 *D. $\frac{\Phi}{\pi d^2} \times \left(\frac{2}{3}\right)^{\frac{3}{2}}$ (40%)

About 40% of the candidates correctly calculated the illuminance at the centre of the floor by applying the inverse square law and summing the contributions from all four sources.

- 3.4 When incident sunlight makes an angle of 20° to the normal of a solar panel, the electric power generated is 100 W.



Assuming that the efficiency of the solar panel remains unchanged, find the electric power generated if the sunlight makes an angle of 30° to the normal of the panel and its intensity is reduced by 10%.

- A. $100 \times \frac{\sin 30^\circ}{\sin 20^\circ} \times 0.1 \text{ W}$ (2%)
 B. $100 \times \frac{\cos 30^\circ}{\cos 20^\circ} \times 0.1 \text{ W}$ (4%)
 C. $100 \times \frac{\sin 30^\circ}{\sin 20^\circ} \times 0.9 \text{ W}$ (13%)
 *D. $100 \times \frac{\cos 30^\circ}{\cos 20^\circ} \times 0.9 \text{ W}$ (81%)

Over 80% of candidates recognised that the effective power received by the panel is proportional to the cosine of the angle of incidence and the intensity of sunlight.

3.8 What are the functions of pressurised water in the core of a pressurised water reactor ?

- (1) It acts as a moderator.
- (2) It acts as a coolant.
- (3) It turns into steam and drives the turbine.

(44%)

*A. (1) and (2) only

(12%)

B. (1) and (3) only

(20%)

C. (2) and (3) only

(24%)

D. (1), (2) and (3)

Candidates who chose options B, C and D wrongly assumed that the pressurised water turns into steam and drives the turbine. In fact, water in a separate secondary loop is turned into steam.

Structured questions

Question 3

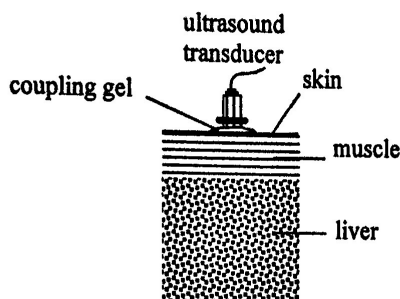
This question assessed candidates' understanding of the energy aspects of lighting and air-conditioning systems. The overall performance of candidates was fair. Part (a)(i) was well attempted, although some confused the concepts of energy efficiency and efficacy. In (a)(ii), less than half recognised that green objects reflect green light and absorb red and blue light. In (b)(i), many applied the correct formula, but just under half substituted the correct values. Some failed to identify that each period lasted 12 hours rather than 24. Performance in (b)(ii) was satisfactory, though a few did not convert the energy unit into kWh. Part (b)(iii) was poorly answered; most did not identify that the minimum power requirement corresponded to the 8 a.m. – 8 p.m. period. The performance in (c) was fair. Some gave suggestions involving changes to the container's dimensions or materials.

Section D : Medical Physics

Multiple-choice questions

Post-examination analysis revealed the following:

4.3 An ultrasound transducer is set up as shown below.



The acoustic impedance of various tissues and the coupling gel used are given below.

	acoustic impedance / $\times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$
coupling gel	1.45
muscle	1.71
liver	1.65

Estimate the proportion of the intensity of the incident ultrasound beam that enters the liver from the coupling gel.

- *A. 0.9929 (43%)
- B. 0.9932 (27%)
- C. 0.9997 (21%)
- D. 0.0071 (9%)

About 40% of candidates correctly calculated the proportion of transmitted intensity using the acoustic impedance values.

4.6 The half-value thickness of a cobalt-60 γ source (1 MeV) is 12.5 mm in lead and 60.5 mm in concrete. Estimate the minimum thickness of a lead sheet to replace a concrete wall of 25.0 cm thick for radiation shielding of this cobalt-60 source.

- A. 6.1 cm (12%)
- *B. 5.2 cm (67%)
- C. 4.0 cm (12%)
- D. 3.1 cm (9%)

Over two-thirds of candidates correctly applied the concept of half-value thickness, recognising that shielding using different materials requires the same number of half-value layers in different materials.

Structured questions

Question 4

This question evaluated candidates' understanding and interpretation of curves of equal loudness. The overall performance was fair. In (a)(i), most candidates failed to recognise that sounds along the same curve are perceived as equally loud by the average person. In (a)(ii), about a third correctly defined the zero-phon curve as the threshold of hearing, while a few mistakenly argued that sounds at this level were inaudible. In (b), about half provided correct answers in both (i) and (ii), though some were unable to describe the curve's features accurately in (ii). The performance in (c) was satisfactory. In (c)(i), about two-thirds correctly determined that loudness and sound intensity levels are equal at 1 kHz. In (c)(ii), more than half successfully calculated the change in sound intensity levels across different frequencies given specific changes in loudness. Candidates performed poorly in (d)(i). Some struggled to apply the threshold of hearing ($1 \times 10^{-12} \text{ W m}^{-2}$) as the reference intensity for determining the ratio. Performance in (d)(ii) was fair. While some recognised that the curves would be raised, they failed to specify the corresponding part of the curve (right end).